1.

Configuration Name	# ADD RS	# DIV RS	# MEM RS	Unified RS
Unified	0	0	0	10
1	2	2	6	0
2	2	6	2	0
3	6	2	2	0
4	2	4	4	0
5	4	2	4	0
6	4	4	2	0
7	3	3	4	0
8	3	4	3	0
9	4	3	3	0
10	2	3	5	0
11	2	5	3	0
12	3	2	5	0
13	3	5	2	0
14	5	2	3	0
15	5	3	2	0
16	4	1	5	0
17	5	1	4	0
18	3	1	6	0
19	6	1	3	0

2.
get_lin_recc

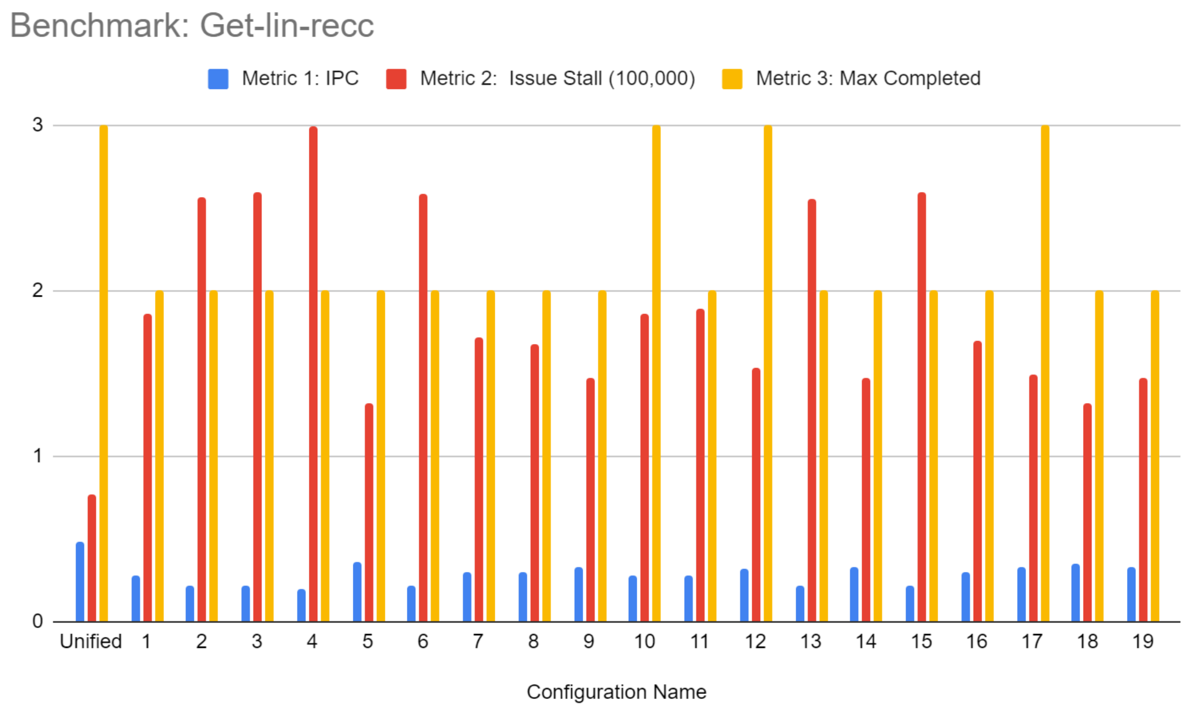
Configuration Name	Metric 1: IPC	Metric 2: Issue Stall	Metric 3: Max Completed
Unified	0.485369	765950	3
1	0.279678	1860590	2
2	0.219917	2562530	2
3	0.217807	2594348	2
4	0.194262	2996334	2
5	0.354134	1317508	2
6	0.218442	2584705	2
7	0.296010	1718073	2
8	0.301210	1675947	2
9	0.329814	1467946	2
10	0.279387	1863283	3
11	0.276670	1888681	2
12	0.319909	1535771	3
13	0.220324	2556456	2
14	0.329816	1467934	2
15	0.217888	2593111	2
16	0.298400	1698523	2
17	0.326663	1489058	3
18	0.353502	1321158	2
19	0.329795	1468075	2

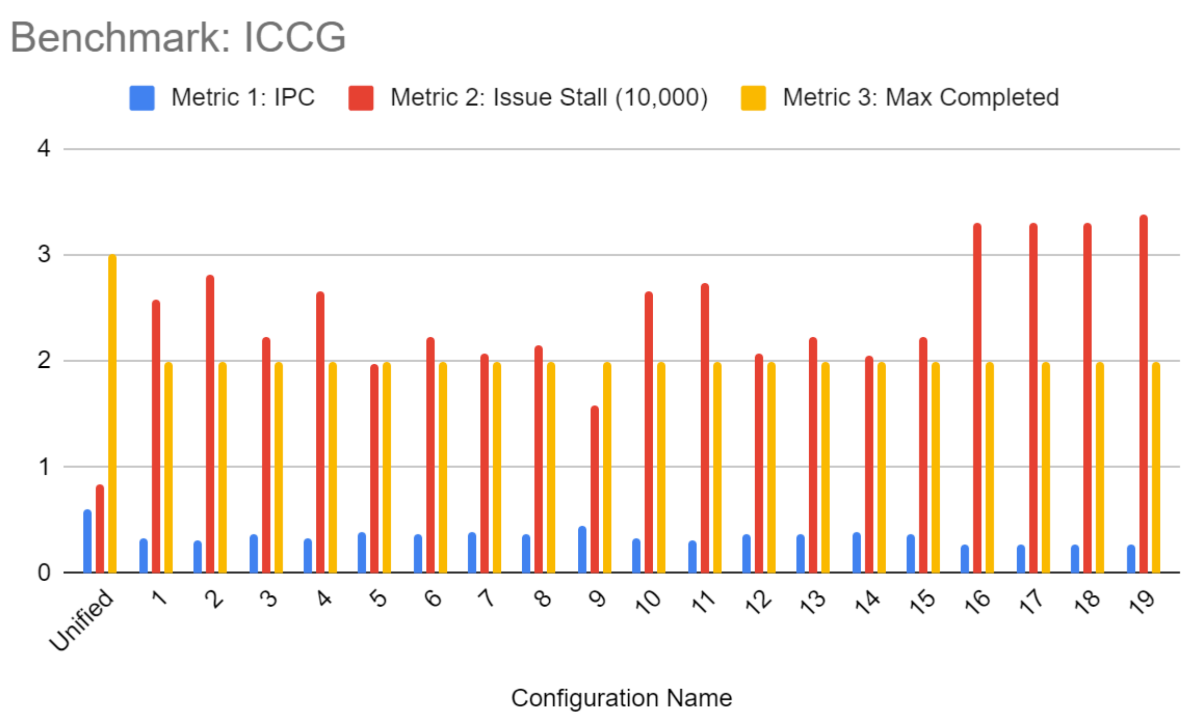
iccg

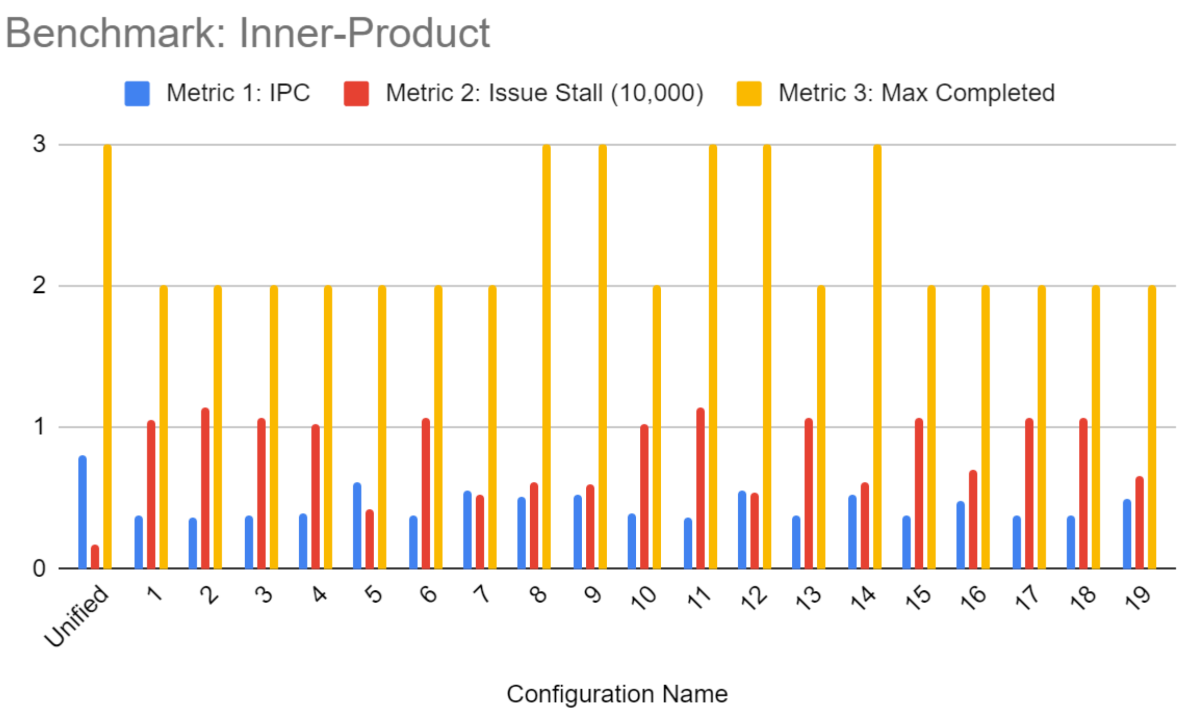
Configuration Name	Metric 1: IPC	Metric 2: Issue Stall	Metric 3: Max Completed
Unified	0.602194	82802	3
1	0.327544	257407	2
2	0.308181	281458	2
3	0.361020	221907	2
4	0.321108	265079	2
5	0.387828	197903	2
6	0.360752	222159	2
7	0.378234	206096	2
8	0.368854	214530	2
9	0.443498	157311	2
10	0.321114	265072	2
11	0.314514	273266	2
12	0.377330	206898	2
13	0.360174	222722	2
14	0.378801	205604	2
15	0.361231	221697	2
16	0.275275	330095	2
17	0.275374	329930	2
18	0.274666	331105	2
19	0.270510	338118	2

inner-product

Configuration Name	Metric 1: IPC	Metric 2: Issue Stall	Metric 3: Max Completed
Unified	0.797820	16585	3
1	0.382921	105565	2
2	0.363313	114783	2
3	0.381185	106327	2
4	0.388652	103026	2
5	0.609239	42006	2
6	0.381108	106362	2
7	0.556587	52162	2
8	0.515508	61542	3
9	0.524463	59372	3
10	0.388654	103025	2
11	0.363541	114668	3
12	0.551202	53313	3
13	0.381037	106394	2
14	0.519498	60581	3
15	0.381108	106362	2
16	0.481127	70641	2
17	0.381086	106379	2
18	0.381028	106405	2
19	0.498224	65968	2


Benchmark: Get-lin-recc
Metric 1: IPC
Metric 2: Issue Stall (100,000)
Metric 3: Max Completed
0
1
2
3
Unified
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
Configuration Name


Benchmark: ICCG
Metric 1: IPC
Metric 2: Issue Stall (10,000)
Metric 3: Max Completed
0
1
2
3
4
Unified
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
Configuration Name



4.
In the three different workloads, the unified reservation station did much better than any of the other combinations of the ADD, DIV, and MEM reservation stations did in terms of getting a higher IPC, also meaning it had the lowest Issue Stall as well. This shows that the issue stall is inversely related to the IPC, so when IPC is higher then the issue stall will be lower, and vice versa. It is also apparent that the average IPC follows this pattern: get-lin-recc < iccg < inner-product. Another observation is that only in get-lin-recc, the issue stall is significantly higher than the other two workloads, where it is in the millions, and the other two are in the 100,000 range. For get-lin-recc, the IPC gets larger as more resources are distributed to the ADD and MEM reservation stations. For iccg, the IPC gets larger as the resources are distributed more evenly among all three reservation stations. And for inner-product, the IPC gets larger as the resources are distributed like a mix of the two others, evenly distributed among all three, but slightly more towards the ADD and MEM reservations stations, shown in configuration name 5.

5.
There could be some reasons such as less data dependency for the workload inner-product that allows for more parallel execution and fewer stalls, letting it complete faster than the other two workloads. This may also be the reason why the unified reservation station was faster, because there may have been issue stalls because there weren't enough reservation stations for that specific instruction, but having a unified one allowed for more to be completed and move on to the next one without having to wait for their specific operation. There could be some instruction scheduling that may affect the efficiency of pipelining, and so if it cannot rearrange the

instruction in such a way, then it may cause more stalls and lower IPCs. in addition the tasks may also affect the results because of different levels of complexity or more resource demands.